Conversion of Camphene to Camphor Day 3

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CHEM 2081 - 007

Introduction

In this experiment, borneol ($C_{10}H_{18}O$) was converted to camphor ($C_{10}H_{16}O$). This experiment was the final day of a three day experiment, with the ultimate goal of converting camphene to camphor. This experiment was performed using a reflux apparatus and water bath.

Methods and Materials

Racemic 0.301 g of borneol product and 0.3 mL of glacial acetic acid (CH_3COOH) were added to a long-neck 10 mL round-bottom flask. This was dissolved in 1 mL of acetone ($(CH_3)_2CO$) and a boiling chip was added. A reflux apparatus was assembled with the distillation column and the apparatus was placed in a water bath with a temperature between 40-50°C. Once the reagents had dissolved, 6 mL of aqueous sodium hypochlorite solution (NaClO) was added through the top opening of the condenser dropwise over 20 minutes. The mixture was gently swirled following each addition. The reaction was allowed to continue for 15 minutes after the last bleach addition. After the mixture cooled to room temperature, the camphor was extracted with two portions (2 mL each) of methylene chloride (DCM) using a centrifuge tube. The combined methylene chloride layers were neutralized with 2 mL of saturated sodium bicarbonate ($NaHCO_3$). Then, the organic solution was washed with 2 mL of 5% sodium bisulfate ($NaHSO_4$) and then with 2 mL of water (H_2O). The organic solution was dried over anhydrous sodium sulfate (Na_2SO_4). The DCM was evaporated over a water bath with a temp of 45°C. The collected crystals were weighed on a watch glass.

Results

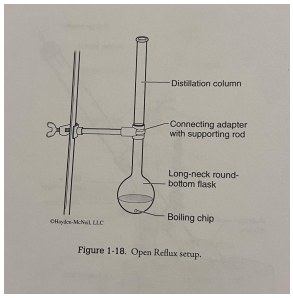


Figure 1-18. Open Reflux setup.

Amount of camphor recovered (g)	0.366

Table 1. Amount of camphor recovered

Calculations

Theoretical yield:

$0.301\ g\ borneol\ \times (1\ mol\ borneol/154.25\ g\ borneol) \times (1\ mol\ camphor/1\ mol\ borneol)$

$\times (152.23\ g\ camphor/1\ mol\ camphor) = 0.297\ g\ camphor$

Actual yield:

$0.366\ g\ camphor$

$(0.366\ g/0.297\ g) \times 100\% = 123.23\%\ yield$

Discussion

In this experiment, 0.366 g camphor was produced from 0.301 g borneol, resulting in a 123.23% yield. Glacial acetic acid and acetone were combined with the borneol product from day 2 and was refluxed. Since the borneol recovered from day 2 was white and powder-like, it is unlikely that there were many, if any, impurities present. Following the reflux and initial extraction, there were two layers present, meaning an organic product was recovered. The layers were light yellow and clear. Since the DCM (1.33 g/mL) has a greater density than water (1 g/mL) it was known that the bottom layer was the organic product. This remained true for the remaining extractions because there were two layers present at each step. One likely source of

error that caused the percent yield to be greater than 100% was if the crystals were still wet when they were weighed. In the final step, the crystals are dried over a water bath. If any water remained in the crystals or on the water bath, this would have caused the percent yield to be greater. To improve this, the crystals should continue to be dried even after the water appears to be gone and the bottom of the watch glass should be dried with a paper towel to remove any water from evaporation. Another source of error was if the water bath during the experiment exceeded 40-50°C. If the water bath was too hot, this could have caused impurities to form during the reflux. This can be improved by keeping a thermometer in the water bath and removing the apparatus if it exceeds 50°C.

References

Department of Chemistry l Auburn University. Conversion of Camphene to Camphor Day 3. In *CHEM 2023 Organic Chemistry II Lab Manual,* 2023 ed.; Macmillan Learning Curriculum Solutions, 2023; pp. 69-70.

Exp. No. 6(D3)	Experiment/Subject	Date
Name	Lab Partner	Locker/ Desk No. / Course & Section No.

- Add racemic 0.3g of your borneol product w/ 0.3mL glacial acetic acid to a long neck 10mL round bottom flask
- Dissolve the racemic mix in 1mL of acetone Add boiling chip
- Assemble reflux w/distillation column & place apparatus in water bath w/temp blw 40-50°C. If borneol doesn't dissolve, add 0.5 mL of acetone until it dissolves
- Add 1mL aq sodium hypochloride sol'n. Add'n should be done through the top opening of the condenser & should be proceed dropwise over 35 min. Gently swirl after each add'n. Allow reaction to continue for 15 min after the last bleach add'n.
- Let mix cool to room temp & extract the camphor w/ 2 portions (2mL each) of methylene chloride. Use centrifuge tube
- Neutralize the combined methylene chloride layers w/ ~2mL of saturated sodium bicarbonate. Presence of acidic residuals will be determined by the production of CO_2 bubbles after every add'n.
- Next, wash the organic sol'n sequentially w/ 2mL of 5% sodium bisulfate & w/ 2mL of water.
- Dry organic sol'n over anhydrous sodium sulfate. It is dry when freely rolling crystals of the drying agent are observed.
- Evaporate the DCM over a water bath w/ temp not exceeding 50°C. Weigh the collected crystals & determine your % yield.

- solvent: water (1 g/mL)
- DCM: 1.33 (g/mL)
- bottom - organic

- 0.301 g
- clear

- done

- done

- light yellow/clear after reflux
- bottom layer - organic

- two layers observed at every step!

- WG: 28.553 g
w/crystals: 28.919 g

SOE: higher than 40-50° still wet when watch glass weighed

Signature	Date	Witness/TA	Date

28.553

Lab Performance

Exp 6 D3

Savannah Creel 007 402 D

Name: *Section:* *Room:* *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ○ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ○ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ○ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ○ ○ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ○ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* [ ]